

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining
(Subclass C)

Date: 13 May, 2024

Time: 6.30pm – 8.30pm

University No:

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This is a closed book examination.

Answer ALL questions.

Write your answers on the exam paper in the corresponding space after the question.

Write your university number on every page and name below.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of calculator: _____

Question 1 (30 marks)

(a) Please provide a concise overview of the main steps in the process of knowledge discovery from data? (5 marks).

(b) Could you kindly provide explanations of the advantages of obtaining latent representations through deep learning, using specific examples from diverse datasets? (5 marks).

(c) Could you kindly outline three distinct data quality issues, providing a concrete real-world example for each problem as an illustration? (5 marks).

(d) Please provide three practical applications and scenarios where graph learning is utilized in real-world contexts, along with a description of the corresponding graph data mining tasks involved? (5 marks).

(e) Could you kindly elaborate on the advantages of large language models in five specific real-world applications? (5 marks).

(f) Could you kindly provide multiple examples of the fundamental methodologies used in handling sequence and graph data, highlighting the key challenges associated with them, and providing a brief overview of the model workflow for each? (5 marks).

Question 2 (40 marks)

(a) We possess a dataset comprising data on individuals' performance in a data mining course (Pass or Fail). The dataset also includes their previous GPA (classified as excellent, average, or poor), study duration (classified as extensive or regular), and assignment scores (classified as below 60, between 60 and 80, or above 80). Our objective is to leverage this dataset to construct a decision tree capable of predicting whether an individual will pass the data mining course based on their GPA, study time, and assignment score. (20 marks).

ID	GPA	Time	Assignment Score	Passed
1	Medium	Normal	60..80	T
2	Low	Much	>80	T
3	High	Much	<60	F
4	High	Much	>80	T
5	Low	Normal	60..80	T
6	Medium	Much	>80	T
7	Low	Normal	<60	F
8	Medium	Much	>80	T
9	High	Normal	<60	T
10	High	Much	60..80	T
11	Low	Normal	>80	F
12	High	Much	>80	T
13	Medium	Normal	<60	F
14	Low	Much	60..80	T
15	Low	Much	<60	F

Q1: What is the entropy $H(\text{Passed})$?

Q2: What is the entropy $H(\text{Passed}|\text{GPA})$?

Q3: What is the entropy $H(\text{Passed}|\text{Time})$?

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Q4: What is the entropy $H(\text{Passed}|\text{Assignment Score})$?

Q5: Please generate the full decision tree that would be learned from this dataset. You do not need to show the calculations.

(b) K-means is a widely utilized clustering algorithm in the field of data analysis. Its primary objective is to group a collection of data points into k distinct clusters, ensuring that each point is assigned to a specific cluster. (20 marks).

Q1. Suppose we have the following data points in two dimensions: $P_1(0, 1)$, $P_2(1, 2)$, $P_3(2, 1)$, $P_4(3, 2)$, $P_5(2, 3)$, $P_6(4, 1)$, and $P_7(6, 6)$, and we want to run the k-means algorithm with $k=2$, using the initial centroids $P_1(0, 1)$ and $P_2(1, 2)$. What would be the final two clusters obtained by running the k-means algorithm? Show step by step the clustering you would obtain by running K-means on the previous set of points.

Q2. If we have a dataset with 1000 data points and want to cluster them into 4 clusters using the k-means algorithm. We randomly initialize the centroids and perform the first iteration, how many distance calculations are needed to compute the new centroids? Please explain your answers.

Q3. Can you provide an example where the k-means algorithm produces at least an empty cluster, meaning the number of non-empty clusters is less than K ? Please provide an example in a one-dimensional Euclidean space with at most 7 points, where $K = 4$.

Question 3 (30 marks)

(a) In modern recommender systems, it is common practice to utilize low-dimensional latent representations for embedding users and items, considering their observed interactions. Furthermore, there is often a wealth of textual information available that is associated with users and items, encompassing detailed item characteristics and customer reviews. Your objective is to develop recommender systems that aim to understand complex user preferences and effectively tackle these challenges by integrating both interaction behavior data (such as clicks, purchases, and likes) and textual data. You are encouraged to explore methodologies like Graph Neural Networks, Large Language Models, and Self-Supervised Learning as potential components of your solution. (15 marks).

(b) Accurate prediction of urban traffic accidents is of importance for the advancement of intelligent transportation systems. It plays a crucial role in enhancing traffic management by providing real-time insights into traffic dynamics, identifying congested areas prone to accidents, and suggesting optimal crowd control measures. Let's consider a scenario where we have comprehensive data on crowd movement across various spatial regions and time intervals. Our objective is to forecast the occurrence of traffic accidents for each region in future time slots. Specifically, imagine a hypothetical scenario where we have accumulated a wealth of data on traffic accidents across various disjoint geographical regions within a city. This dataset covers a specific duration, comprising multiple time slots, such as 30 minutes or an hour each. Our primary objective is to create a predictive model that can effectively forecast the traffic volume for each region in the subsequent time slot, denoted as $(T+1)$. As you embark on this task, it is vital to acknowledge the technical challenges posed by the intricate spatial and temporal correlations inherent in accident prediction. (15 marks).